

6L Live PlmCasX Expression & Purification

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Day 1:

- Transform plasmid into competent Rosetta cells (do entire step by Bunsen burner)
 - Thaw 10uL competent Rosetta cells on ice, flick to mix
 - Add 1uL ~200uM plasmid to cells, rest on ice 5-10 min
 - Heat shock cells in 42 C water bath, 35-45 s
 - Add 300uL room temp LB media, rest on ice 2 min
 - Shake cells 160-180 rpm, 37 C, 1 hr
- OR take stab from -80 C glycerol stock
- Grow pre-culture
 - Add 200mL LB to 500mL baffled Erlenmeyer flask
 - Add amp to a final concentration of 50 mg/L
 - 200uL 50mg/mL amp
 - Add entire tube of transformed cells to flask
 - Shake 160-180 rpm, 37 C, overnight

Day 2:

- Grow TB culture (do entire step near Bunsen burner)
 - Locate 6, 1 L TB in glass or plastic baffled flasks
 - Add amp to a final concentration of 50 mg/mL
 - 1 mL 50 mg/mL amp per flask
 - Add 20-25 mL pre-culture in LB to each flask of TB
 - Shake 160-180 rpm, 37 C, 3-6 hrs
 - Grow until OD600 is 0.5-0.6
 - Make sure to blank spectrophotometer with cuvette of clean media
- Induce protein expression
 - Transfer 1 L TB flasks to ice
 - Add 1 mM IPTG final to each flask
 - 1 mL 1 M IPTG per flask
 - Shake 160-180 rpm, 16 C, overnight

Day 3:

- Lyse cells
 - Transfer cultures to 1 L plastic centrifuge tubes
 - Balance pairs of 1 L tubes by adding MiliQ H₂O
 - Centrifuge cultures in Beckman Coulter J6-MI
 - Use blue sleeves to ensure tubes fit snugly
 - 4,000 rpm, 20 min, 4 C
 - Pour off supernatant from cultures
 - Add 20 mL Lysis Buffer + TCEP to each plastic tube
 - Vortex tubes until sample is homogenous
 - Use flat top vortex

- Transfer cell suspension to 500 mL beaker
 - Alternatively, lysed cells can be frozen down for later use. Transfer to 5 mL tubes. Centrifuge 4,000 rpm, 20 min, 4 C. Pour off supernatant. Freeze at -20 C
- In separate beaker, add 10 mL Lysis Buffer + TCEP
 - Add 8 tablets ROCHE cOmplete tablets
 - 1 tablet per 20 mL
 - Add PMSF to 0.5 mM final
 - 800 uL 100 mM stock PMSF
 - Add 8 uL RNase free DNase
 - 2uL per 20 mL
- Add solution of tablets + PMSF + DNase to cell suspension
- Place sample in tube of ice w/ water to hold beaker snugly
- Sonicate sample in cold room
 - Rinse tip of sonicator w/ EtOH then H₂O
 - Set sonicator to 5 min, 10 s on, 15 s off, 5.5 power setting
- Place 6 plastic ultracentrifuge tubes w/ cap on ice
- Divide sonicated sample between tubes
 - Use scale to match pairs withint +/- 0.01g by adding MiliQ H₂O
- Centrifuge samples in Beckman Coulter Avanti J-E
 - 35,000 rpm, 45 min, 4 C
- Filter sonicated sample through bottle top 0.22 um filter, store on ice
- Ni-NTA column binding
 - Add 20 mL Qiagen Ni-NTA resin to yellow gravity column with butterfly valve in 4 C cold room
 - Drain off EtOH storage buffer
 - Wash w/ 10 mL Lysis Buffer + TCEP, repeat 5x
 - Add 10 mL Lysis Buffer + TCEP to Ni-NTA column, mix with pipette
 - Transfer Ni-NTA resin to bottle with sonicated/filtered sample, add stir bar
 - Stir sample + Ni-NTA resin in 4 C cold room, at least 2 hrs
 - Add sample + Ni-NTA resin back to yellow gravity column
 - Drain liquid
 - Collect “flow through” sample for gel
 - Add 50 mL Lysis Buffer + TCEP, drain off
 - Collect “wash” sample for gel
 - Add 10 mL Lysis Buffer + TCEP
 - Add TEV protease
 - 4 eppendorfs of high conc. Lab stock
 - Mix sample + Ni-NTA resin well
 - Incubate column 4 C, overnight

Day 4:

- Elution of Ni-NTA column
 - Elute Ni-NTA column into 50 mL tube
 - Keep on ice
 - Label as “flow through”
 - Wash column with 10mL Elution Buffer (500 mM imidazole), 3x

- Collect samples into 50 mL tubes
 - Label as "Elution #)
- Pool fractions into 15 mL 50 kDa MWKO Amicon Concentrators
- Centrifuge 4,000 rpm, 4 C
 - Centrifuge until volume is < 4 mL
- Transfer to 4 mL 30 kDa MWKO Amicon Concentrator
 - Centrifuge until volume is < 500 uL
- SP Column Ion Exchange
 - Run Akta at 0.25 mL/min to attach column
 - 5 mL SP Column
 - Screw bottom of column into top Akta port
 - Make sure to use wet connections
 - 0.5 MPa, Pre-column pressure alarm
 - Equilibrate column with 40 mL IEX A Buffer
 - 5-10 CVs
 - Run at 1-5 mL/min
 - 0.5 MPa pre-column pressure alarm
 - Run Akta at 0.25 mL/min to detach column
 - Draw up 500 uL sample in 1 mL syringe
 - Attach threaded luer lock attachment
 - Remove air from syringe
 - Manually inject sample into top of 5 mL SP column
 - Use wet connections
 - Don't exceed 5 mL/min
 - Run Akta at 0.25 mL/min to attach column
 - Insert new 96 deep well plate into position 3
 - Run SP column gradient program between IEX Buffer A and B to elute
 - Collect fractions from chromatogram peak
 - Save samples for gel
 - Concentrate fractions if necessary

Day 5:

- Size Exclusion Chromatography
 - Run Akta at 0.25 mL/min to attach S200 10/300 increase column
 - 0.5 mL/min
 - 2.8 MPa pre-column pressure
 - 24 mL column
 - Add 500 uL loop to Akta
 - Inject 3 mL SEC Buffer into loop with 5 mL syringe + red luer lock attachment
 - Draw up sample into 1 mL syringe + red luer lock attachment
 - Inject sample (should be < 500 uL) into loop, leaving syringe attached
 - Run SEC program
 - Should inject sample from loop
 - Collect fractions from chromatogram
 - Save samples for gel
 - Concentrate as necessary

- Nanodrop protein, aliquot into tubes, flash freeze in LNO₂, store at -80 C